



A flexible model for spatio-temporal modeling of low-threshold exceedances

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Outline

This work develops a spatio-temporal framework for modeling low-threshold exceedances based on extreme value theory (EVT), establishes its tail properties, and demonstrates its performance through simulation studies and a real-data application.

Introduction

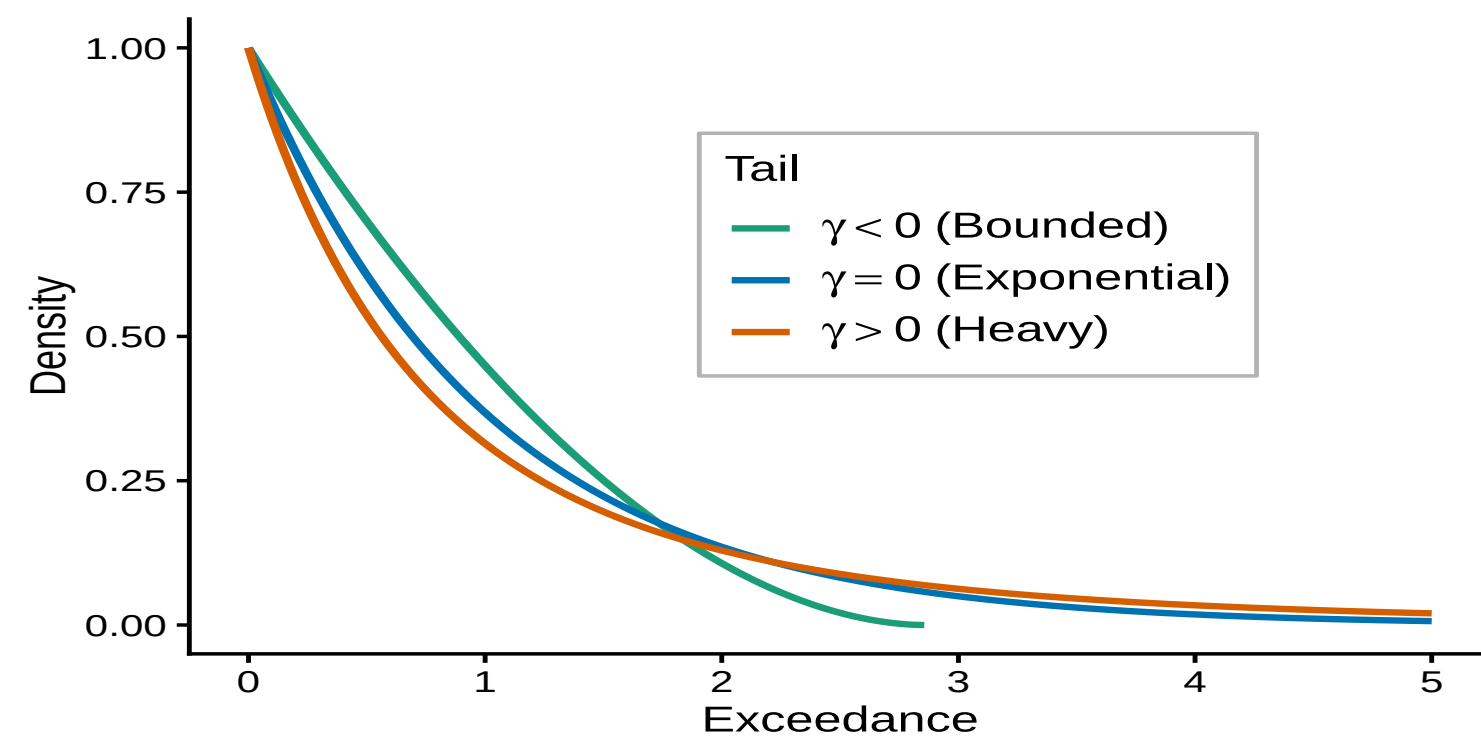
Modeling environmental extremes, such as heavy rain-fall, extreme temperatures, and wind speed gusts, is fundamental to climate science, hydrology, and risk assessment. Peak-over-threshold approach models upper-tail behavior by generalized Pareto distribution (GPD) for sufficiently high threshold exceedances. However, the GPD performs poorly at low thresholds, leading to biased parameter estimates. Addressing this limitation while preserving tail behavior remains an important challenge.

Background

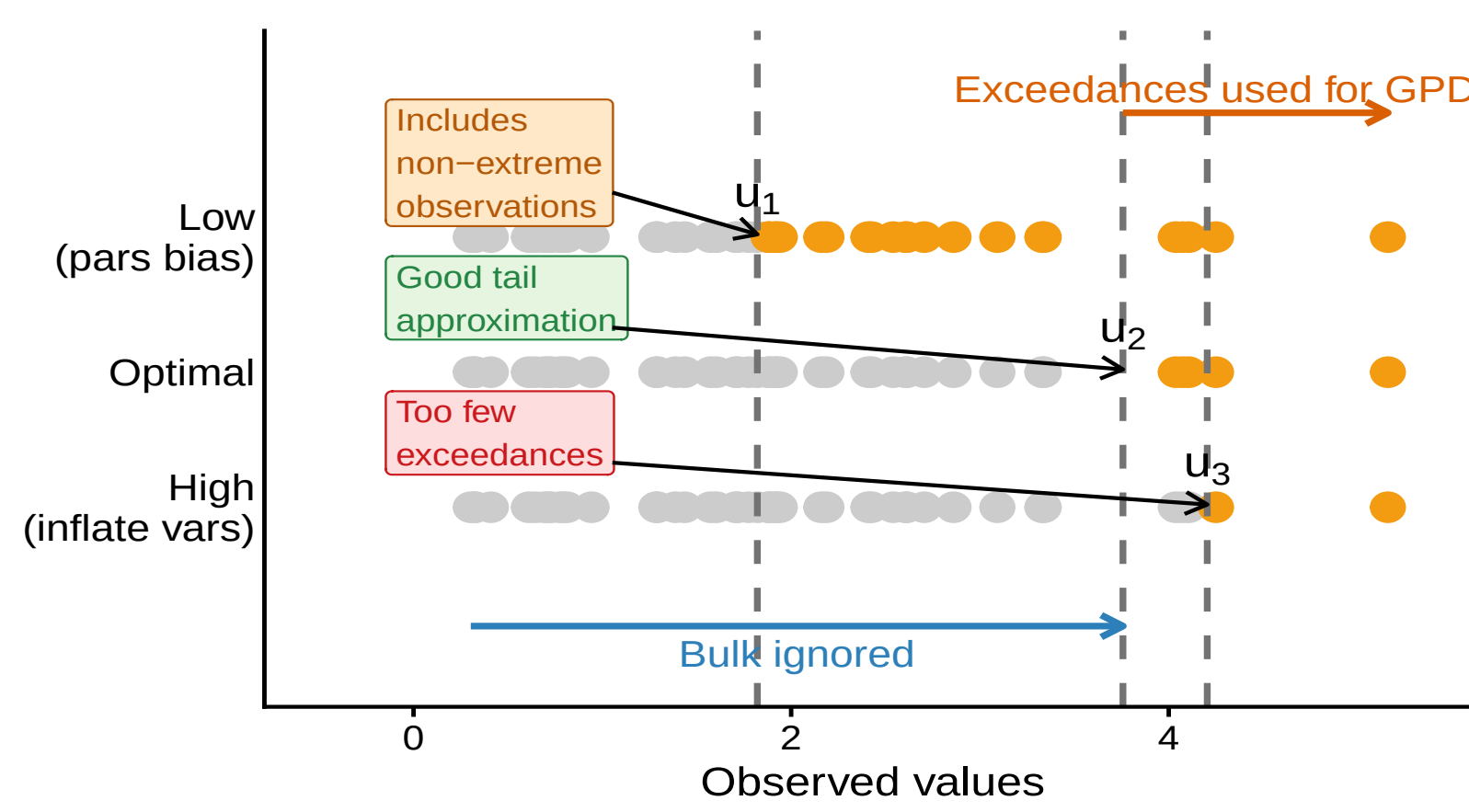
Let Y be a non-negative random variable with distribution function F and right endpoint $y^F := \sup\{y : F(y) < 1\} \in (0, \infty]$. Positive exceedances above a high threshold u are modeled by the GPD

$$\mathcal{H}(y; \boldsymbol{\theta}) = 1 - \left(1 + \frac{\gamma y}{\sigma}\right)_+^{-1/\gamma}, \quad h(y; \boldsymbol{\theta}) = \frac{1}{\sigma} \left(1 + \frac{\gamma y}{\sigma}\right)_+^{-1/\gamma - 1}, \quad (1)$$

with $\boldsymbol{\theta} = (\sigma, \gamma) \in \mathbb{R}^+ \times \mathbb{R}$. How does the GPD tail behave?



Problem: GPD is asymptotic in the threshold u and provides no information about the bulk region $\{Y \leq u\}$. In finite samples, the estimator $\hat{\gamma}(u)$ is subject to the classical bias-variance dilemma: increasing u reduces the approximation bias in (1) but discards data and inflates variance.



Proposed extended GPD (eGPD)

To break mean-variance trade-off, we can GPD probability integral transform

$$Y = \mathcal{H}^{-1}(W; \boldsymbol{\theta}), \quad W \sim K_W(\cdot; \boldsymbol{\psi}), \quad \sup W \subseteq [0, 1], \quad (2)$$

which transforms the bulk through K_W . For $y > 0$,

$$\mathcal{F}(y) = K_W(\mathcal{H}(y; \sigma, \gamma); \boldsymbol{\psi}), \quad (3)$$

For a K_W , we need to satisfy the following conditions (Naveau et al., 2016; Gamet and Jalbert, 2022):

- C_1 : $\lim_{w \uparrow 1} k_W(w; \boldsymbol{\psi}) = a$ (upper-tail matching),
- C_2 : $k_W(0; \boldsymbol{\psi}) > 0$ (positive density at the origin),
- C_3 : $\lim_{w \uparrow 1} k'_W(w; \boldsymbol{\psi}) = 0$ (first-order tail smoothness).

Generator for K_W : For $\kappa > 0$, let W have support $[0, 1]$, mode at $w = 1$, scale $s = \kappa^{-1}$, and

$$K_W(w; \kappa) = \frac{1}{Z} \left(\frac{1}{1 + e^{-\kappa(w-1)}} - \frac{1}{1 + e^\kappa} \right), \quad (4)$$

where $Z := \frac{1}{2} - (1 + e^\kappa)^{-1} > 0$ normalizes K_W on $[0, 1]$.

Conditions K_W satisfied: For every $\kappa, Z > 0$ C_1 . $k_W(1; \kappa) = \frac{\kappa}{4Z} > 0$, C_2 . $k_W(0; \kappa) = \frac{\kappa}{Z} e^\kappa (1 + e^\kappa)^{-2} > 0$ and C_3 . After some algebra as $w \uparrow 1$, $u \rightarrow 1$, so $\lim_{w \uparrow 1} k'_W(w; \kappa) = 0$.

Substituting (4) into (3) yields eGPD, with parameters $\boldsymbol{\theta} = (\sigma, \gamma, \kappa) \in \mathbb{R}^+ \times \mathbb{R} \times \mathbb{R}^+$,

$$\mathcal{F}(y; \sigma, \gamma, \kappa) = \frac{1}{Z} \left(\frac{1}{1 + e^{-\kappa(\mathcal{H}(y)-1)}} - \frac{1}{1 + e^\kappa} \right). \quad (5)$$

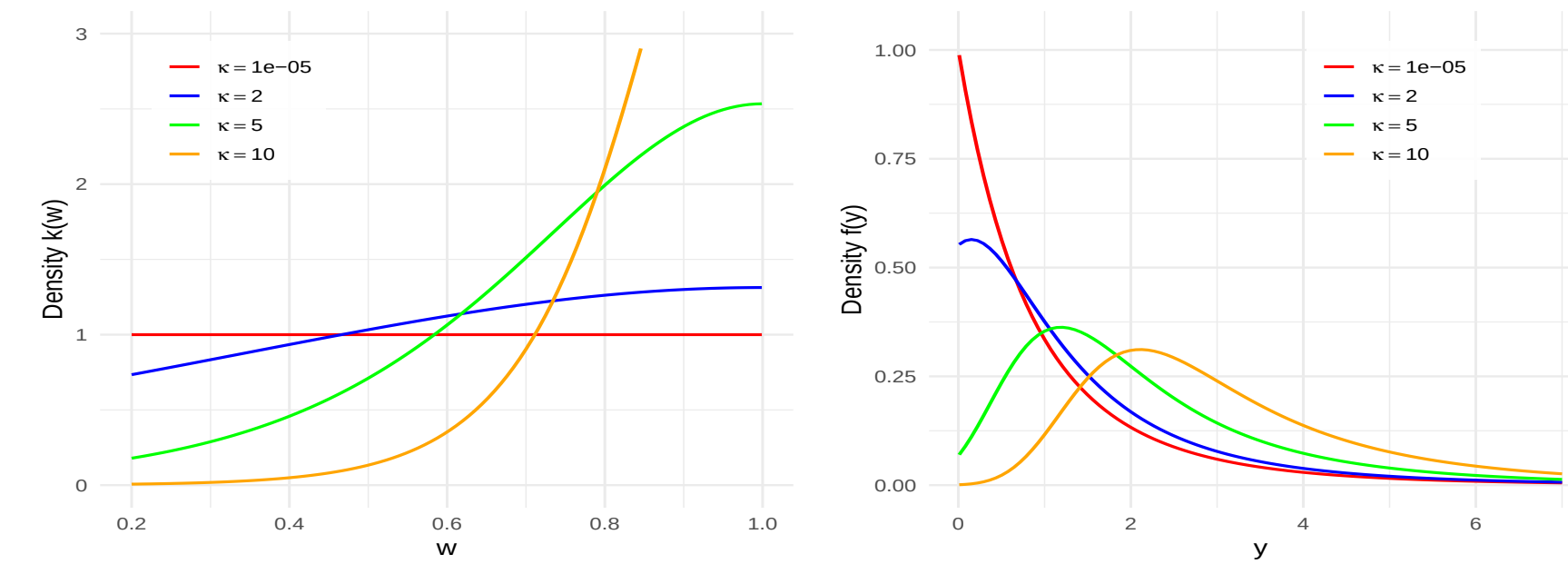


Figure 1. (a) Density of K_W . (b) eGPD for fixed $\sigma = 1$ and tail index $\gamma = 0.2$.

Generlized Additive Model (GAM) Setting

Let $Y_t(s)$ denote the realizations of the process at the location $s \in S$ and discrete time $t \in T = \{1, \dots, T\}$. For a low threshold $u_t(s)$ we assume that

$$Y_t(s) - u_t(s) \mid Y_t(s) > u_t(s) \sim \text{eGPD}(\kappa_t(s), \sigma_t(s), \gamma_t(s)).$$

The parameters $\kappa_t(s)$, $\sigma_t(s)$, and $\gamma_t(s)$ are represented via link functions and linear predictors in the covariates.

For selection $u_t(s)$, we use quantile regression

$$Y_t(s) \mid u_t(s) \sim \text{ALD}(u_t(s), \psi_t(s)), \quad (6)$$

where ALD is the asymmetric Laplace distribution. The general GAM formulation may be used to represent a parameter $\Psi_t(s)$ as

$$h(\Psi_t(s)) = \beta_0 + \sum_{j=1}^g \sum_{d=1}^{D_j} \beta_{jd} b_{jd}(s, t), \quad (7)$$

where $h(\cdot)$ denotes a suitable link function.

Simulation study

Simulation-based evaluation of eGPD.

Algorithm 1 Simulation study for spatial return level estimation

- 1: **Input:** Spatial domain S , test locations $S_{\text{test}} \subset S$, simulation length $T_{\text{sim}} = 10,950$ time points across $J_{\text{sim}} = 100$.
- 2: **Output:** Estimated 100-year return levels $\hat{x}_{100}(s)$.
- 3: Generate locations S_{sim} from a homogeneous Poisson process.
- 4: Simulate GP for the Weibull parameters $\alpha(s)$ and $\beta(s)$.
- 5: Compute the true 100-year return level $x_{100}(s)$ at each $s \in S_{\text{test}}$.
- 6: **for each** $s \in S_{\text{sim}}$ **do**
- 7: Generate T_{sim} Weibull observations.
- 8: **end for**
- 9: Estimate spatial thresholds $\hat{u}(s)$ using quantile regression.
- 10: Fit the eGPD and GPD models to threshold exceedances.
- 11: **for each** $s \in S_{\text{test}}$ **do**
- 12: Estimate $\hat{x}_{100}(s)$ using the fitted models.
- 13: **end for**

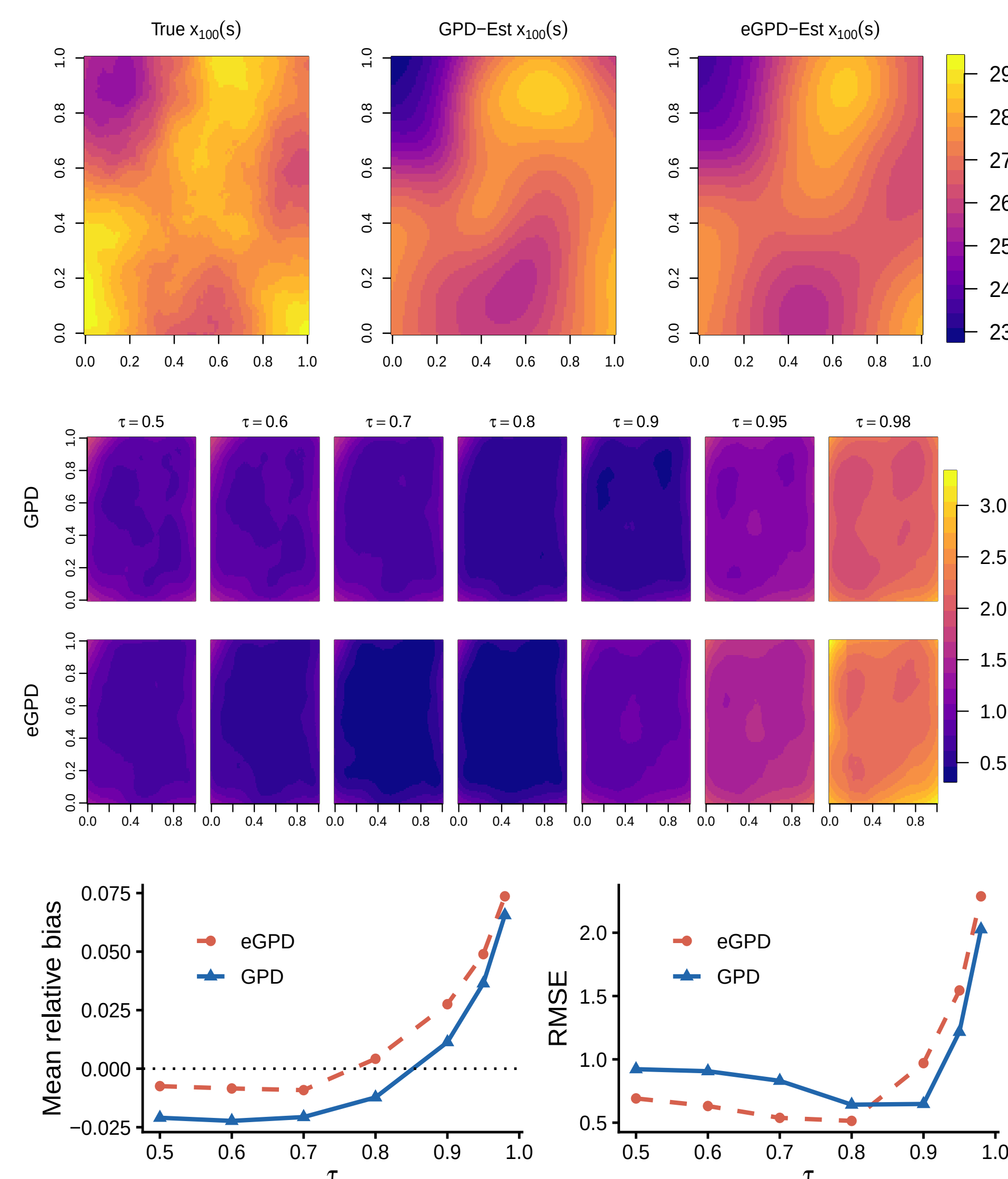


Figure 2. Simulation results comparing the GPD and eGPD. Top: true versus estimated return levels at $\tau = 0.50$. Middle: spatial RMSE across τ . Bottom: spatially aggregated RMSE across τ .

Real data application

Daily maximum US wind gust speeds (1997–2016) from stations in the NCDC GSOD database.

$$\begin{aligned} \log \sigma(s, t) &= \tilde{g}(g_{\sigma, \text{tp}, 40}(s), g_{\sigma, \text{cc}, 5}(t_{\text{cyc}})) + g_{\sigma, \text{nc}, 15}(\text{mws}(s)), \\ u(s, t) &= \tilde{g}(g_{u, \text{tp}, 100}(s), g_{u, \text{cc}, 6}(t_{\text{cyc}})) + g_{u, \text{cr}, 15}(\text{mws}(s)), \\ \gamma(s, t) &= \tilde{g}(g_{\gamma, \text{tp}, 30}(s), g_{\gamma, \text{cc}, 5}(t_{\text{cyc}})), \\ \kappa(s, t) &= \tilde{g}(g_{\kappa, \text{tp}, 30}(s), g_{\kappa, \text{cc}, 5}(t_{\text{cyc}})). \end{aligned}$$

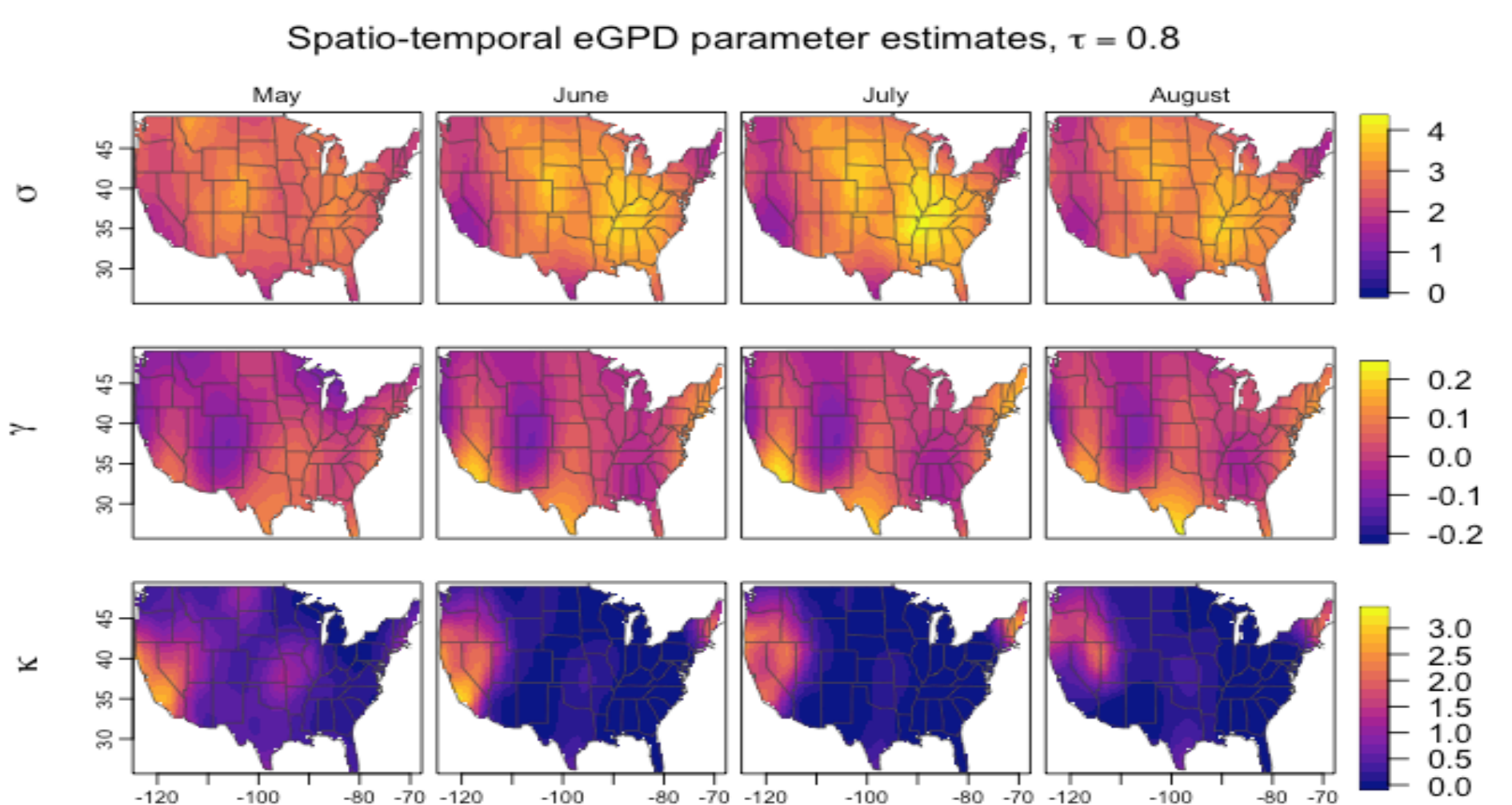
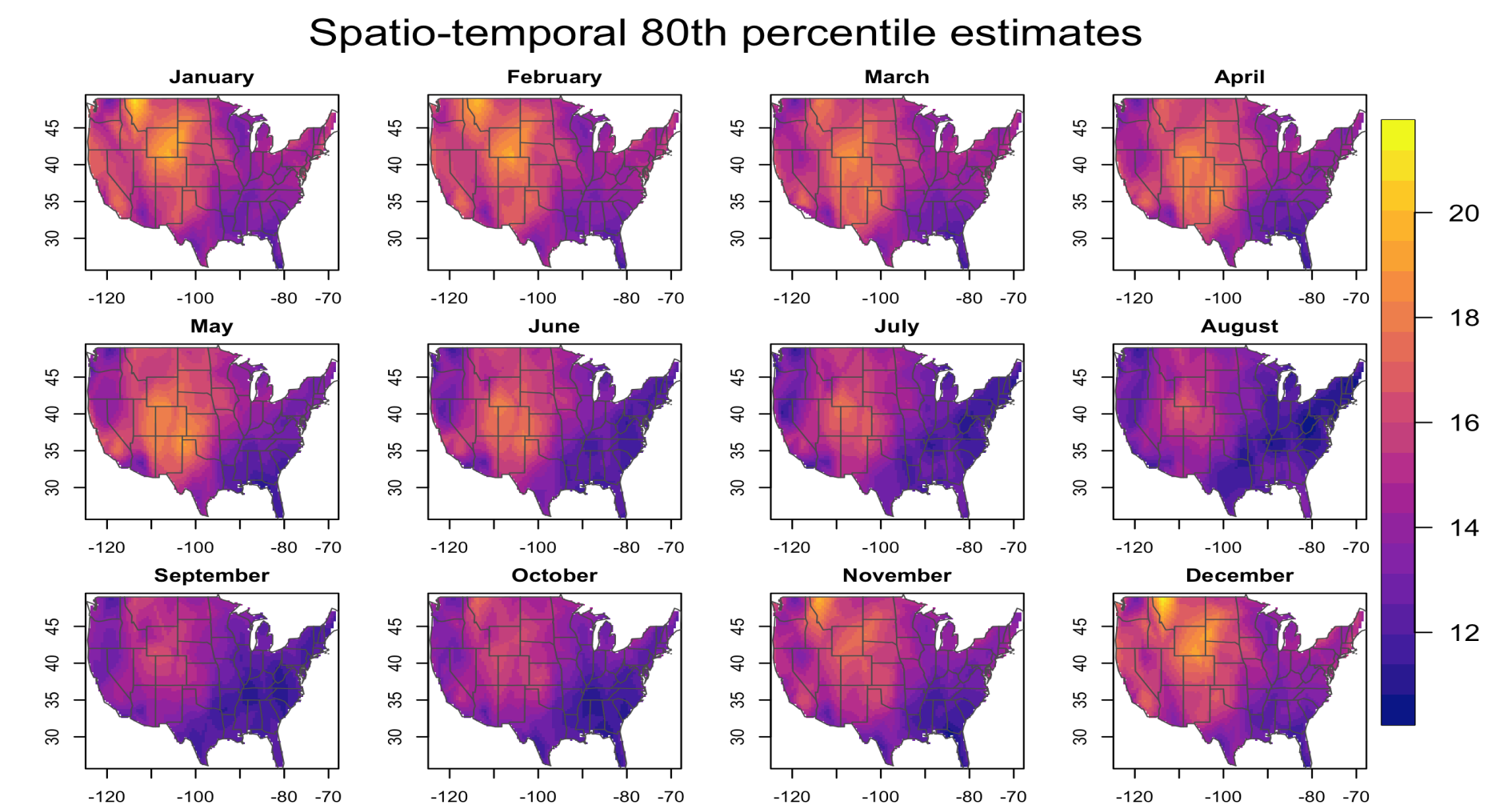


Figure 3. Real data analysis for the U.S. wind gusts application. From top to bottom: estimated threshold surface $u_t(s)$ at the quantile level $\tau = 0.80$; spatial estimates of the eGPD parameters; ΔAIC comparison between the GPD and eGPDs across candidate threshold quantiles; and estimated return levels based on the fitted GPD and eGPD model and their difference at $\tau = 0.80$.

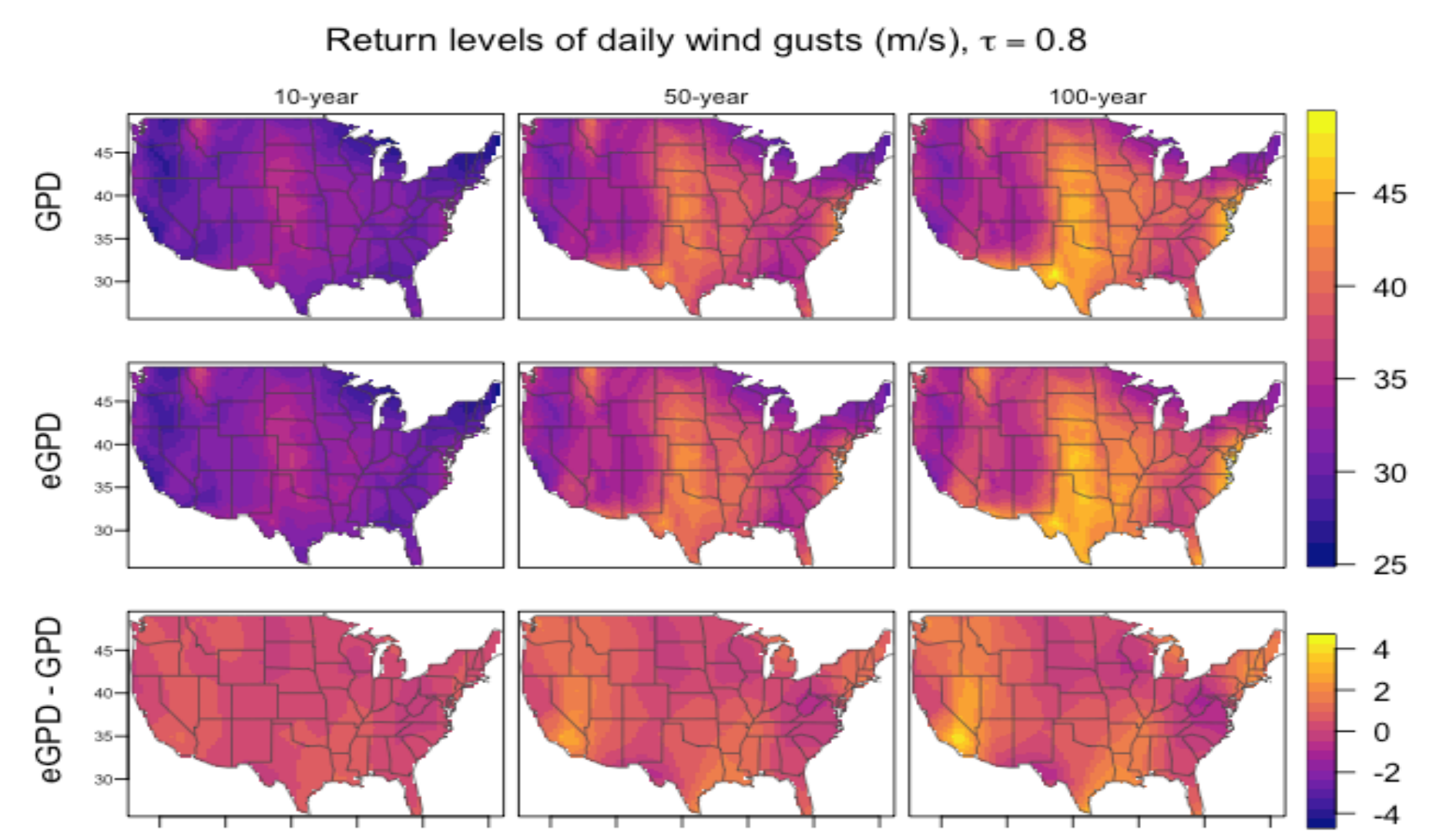
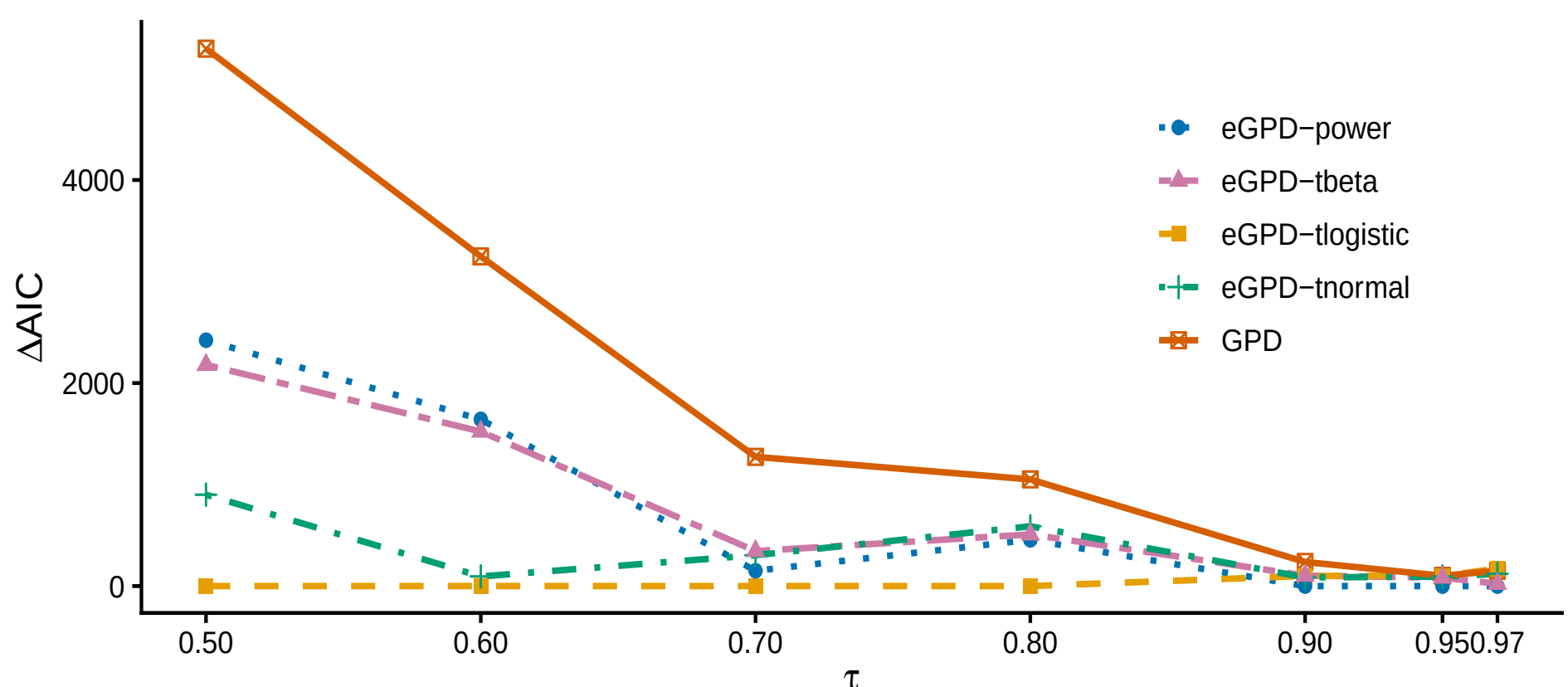


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Conclusion and Future work

The proposed model consistently achieves superior performance for low threshold exceedances. We are also interested in extending its use to various applications while explicitly accounting for spatial dependence.

Acknowledgement

This work has received funding from the European Union's Horizon Europe research and innovation programme under the Marie Skłodowska-Curie grant agreement No. 101126636.

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